

Notes: Petrin (JPE, 2002)

Quantifying the Benefits of New Products: The Case of the Minivan

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1 Big picture

- **Question.** How large were the consumer and producer welfare effects of the introduction of the minivan, and how should one estimate these effects when only market-level product data are available?
- **Setting.** The paper studies the U.S. automobile market from 1981 to 1993. Chrysler introduced the Dodge Caravan minivan in 1984; Ford and General Motors later entered with their own minivans.
- **Core empirical challenge.** Welfare from a new differentiated product depends on substitution patterns. If the model thinks consumers substitute randomly across products, or relies heavily on an idiosyncratic logit taste shock, it can greatly overstate willingness to pay for the new product.
- **Main methodological contribution.** The paper extends the Berry, Levinsohn, and Pakes random-coefficients discrete-choice framework by adding *micro moments*: moments that relate average buyer demographics to the characteristics of the products they buy.
- **Why micro moments matter.** Petrin does not need fully linked consumer-level choice data for every product. Instead, he uses aggregate Consumer Expenditure Survey information, such as average family size of minivan buyers and new-vehicle purchase probabilities by income group, to discipline taste heterogeneity.
- **Main demand insight.** Family size helps identify tastes for minivans, station wagons, sport-utility vehicles, and full-size vans; income groups help identify price sensitivity and income effects.
- **Main welfare result.** Models without micro moments generate much larger compensating-variation estimates. For 1984 minivan buyers, average compensating variation falls from \$7,414 in the IV logit model and \$3,171 in the random-coefficients model without micro moments to \$1,247 in the random-coefficients model with micro moments.
- **Main diagnostic result.** The models without micro moments rely too much on the idiosyncratic logit taste term. With micro moments, welfare is driven more by observed characteristics and less by an unexplained “love of variety” shock.
- **Producer-side result.** Chrysler gained substantially from being first mover. In 1984, the minivan increased Chrysler’s estimated variable profits by about \$202.5 million, while Ford and GM lost profits because minivans cannibalized their existing products.

- **Total welfare result.** Over 1984–1988, the paper estimates roughly \$2.8 billion in consumer surplus gains and about \$105 million in producer-profit gains, for total welfare gains of about \$2.91 billion.
- **Economic takeaway.** New products can create large consumer benefits, but credible measurement requires realistic substitution patterns. Competition among firms can generate consumer welfare by creating differentiated products, even when the innovation cannibalizes rivals' profits.

2 Data and measurement (key objects)

2.1 Why the minivan setting is useful

- The minivan was a major new product in an important durable-goods market.
- It had clear substitutes: station wagons, large sedans, sport-utility vehicles, and full-size vans.
- The historical narrative is economically sharp: Ford considered a minivan-like vehicle earlier but worried about cannibalizing its station wagon sales; Chrysler introduced the minivan in 1984 and gained a temporary first-mover advantage.
- The welfare question is meaningful because the product was genuinely new, not merely a small quality improvement to an existing nameplate.
- The setting is also useful for studying producer effects because the Big Three competed in overlapping product spaces, so Chrysler's innovation likely shifted both consumer surplus and rival profits.

Interpretation. The minivan case is a clean example of a broad economic question: how much value do consumers receive when a firm introduces a product that occupies a new point in characteristics space?

2.2 Market-level automobile data

- The market data contain all new vehicle nameplates sold in the United States from 1981 to 1993 with sales above 1,000 vehicles.
- The sample includes 2,407 vehicle-year observations.
- Data sources are *Automotive News Market Data Book* and *Ward's Automotive Yearbook*.
- Vehicle characteristics include horsepower divided by weight, size, fuel efficiency, front-wheel drive, air conditioning as standard equipment, vehicle type indicators, price, and quantity sold.
- The paper treats nameplate-year observations in adjacent years as the same model when horsepower, wheelbase, width, and length do not change by more than 10%. Under this definition, there are 916 independent observations.

Interpretation. The market-level data provide the standard BLP-style ingredients: product shares, prices, observed characteristics, and firm ownership. These data identify average demand and supply moments, but they are often weak for identifying detailed substitution patterns.

2.3 Consumer Expenditure Survey data

- Consumer demographics come from the Consumer Expenditure Survey (CEX) for 1987–1992.
- The main CEX sample contains nearly 30,000 household observations and is used to approximate the empirical distribution of U.S. household demographics.
- The CEX automobile supplement reports new vehicle purchases and links buyers to demographics.
- The paper observes 2,660 new vehicle purchases in the supplement.
- For the key family-vehicle categories, the sample sizes are limited but useful: 120 minivan buyers, 63 station wagon buyers, 131 sport-utility buyers, and 23 full-size van buyers.

Interpretation. The CEX supplement does not provide enough observations to estimate a full consumer-level choice model directly. Its value is that it provides aggregate demographic moments that can discipline the aggregate demand model.

2.4 Product characteristics

The main observed product characteristics are:

- **Price.** Base-model list price, measured in 1982–1984 CPI-adjusted dollars.
- **Acceleration.** Horsepower divided by curb weight.
- **Size.** A physical vehicle-size measure based on dimensions.
- **Fuel efficiency.** Miles per dollar.
- **Luxury proxy.** Indicator for air conditioning being standard.
- **Front-wheel drive.** Indicator for front-wheel drive.
- **Family vehicle type.** Indicators for minivans, station wagons, sport-utility vehicles, and full-size passenger vans.

Interpretation. These characteristics are the dimensions along which the model allows households to substitute. If large families especially value interior space or minivan-like attributes, the model needs to estimate that pattern instead of assigning it to an unexplained product fixed effect.

2.5 Micro moments

The micro moments summarize demographic patterns among new-vehicle buyers. They include:

- probabilities of buying a new vehicle for low-, middle-, and high-income households;
- average family size among buyers of minivans, station wagons, sport-utility vehicles, and full-size vans;
- probability that the household head is between ages 30 and 60 among buyers of those family-vehicle categories.

Interpretation. These moments tell the model who buys what. For example, the fact that minivan buyers have larger families helps identify a family-size taste shifter for minivans and related family vehicles.

2.6 Descriptive facts

- Minivans and station wagons are close in customer base and product function.
- Minivan buyers have larger families than the average new-vehicle buyer.
- Station wagon sales fall as minivan sales rise, suggesting that minivans cannibalized a major pre-existing family vehicle segment.
- Chrysler's minivan was differentiated because it had front-wheel drive and a passenger-car-like platform, while Ford and GM initially responded with more van-like products built on truck or van platforms.

Interpretation. The descriptive evidence already points to substitution between minivans and other family vehicles. The structural model quantifies how much this substitution matters for welfare and markups.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic structure

The model combines a differentiated-products demand system with a supply-side pricing model.

- Consumers choose one product from the available vehicle set or choose the outside option.
- Products differ in observed characteristics, prices, and unobserved characteristics.
- Consumers differ in income, family size, age, and unobserved tastes.
- Firms are multiproduct oligopolists that set prices in a static Bertrand-Nash equilibrium.
- Counterfactual welfare is computed by removing minivans from the choice set and solving for the new equilibrium prices of remaining vehicles.

Interpretation. The paper needs both demand and supply. Demand tells us which products consumers switch to and how much they value the minivan. Supply tells us how prices and producer profits change when the new product enters or exits.

3.2 Indirect utility

Consumer i 's utility from product j is

$$u_{ij} = \alpha_i \ln(y_i - p_j) + X_j \beta + \sum_k \gamma_k \nu_{ik} x_{jk} + \xi_j + \varepsilon_{ij}.$$

Here:

- y_i is household income;
- p_j is product price;
- X_j is the vector of observed vehicle characteristics;
- β is the vector of mean taste parameters;
- ν_{ik} is consumer i 's idiosyncratic taste draw for characteristic k ;
- γ_k controls heterogeneity in taste for characteristic k ;
- ξ_j is the product characteristic observed by firms and consumers but not by the econometrician;
- ε_{ij} is the idiosyncratic type I extreme value logit taste shock.

Interpretation. The first key extension beyond simple logit is that households differ in tastes for observed characteristics. The second key extension is that the paper uses micro moments to discipline these heterogeneous tastes.

3.3 Income-group price sensitivity

The marginal utility of income is allowed to differ across three income groups:

$$\alpha_i = \begin{cases} \alpha_0, & y_i \leq \bar{y}_1, \\ \alpha_1, & \bar{y}_1 < y_i < \bar{y}_2, \\ \alpha_2, & y_i \geq \bar{y}_2, \end{cases}$$

where $\bar{y}_1$ and $\bar{y}_2$ split the U.S. population into three equally sized income groups.

Interpretation. New-vehicle purchase probabilities differ sharply by income. Matching purchase rates by income group helps identify how price sensitivity varies across households.

3.4 Family-size taste interactions

For family vehicles, taste heterogeneity is allowed to depend on household family size. For example, for minivans and station wagons:

$$\gamma_{i,mi} = \gamma_{mi} \ln(f s_i) \nu_{i,fv}, \quad \gamma_{i,sw} = \gamma_{sw} \ln(f s_i) \nu_{i,fv},$$

where $f s_i$ is family size and $\nu_{i,fv}$ is a common idiosyncratic family-vehicle taste draw.

Analogous terms are included for sport-utility vehicles and full-size passenger vans.

Interpretation. Larger families can have systematically stronger tastes for family vehicles. The common family-vehicle taste draw also allows these products to be closer substitutes for each other.

3.5 Choice probabilities and market shares

After integrating out the type I extreme value taste shock, the probability that household i chooses product j is

$$P_{ij} = \frac{\exp[\alpha_i \ln(y_i - p_j) + X_j \beta + \sum_k \gamma_k \nu_{ik} x_{jk} + \xi_j]}{\sum_\ell \exp[\alpha_i \ln(y_i - p_\ell) + X_\ell \beta + \sum_k \gamma_k \nu_{ik} x_{\ell k} + \xi_\ell]}.$$

Aggregate market share is obtained by integrating over the empirical distribution of demographics and over simulated unobserved taste draws:

$$s_j(p, X; \theta) = \int P_{ij}(p, X; \theta) dF(i).$$

Interpretation. The model predicts both product market shares and demographic composition of buyers. The estimator chooses parameters so that both sets of predictions resemble the data.

3.6 Demand inversion

As in BLP, for a candidate vector of nonlinear parameters, the mean utility terms are chosen so that predicted shares equal observed shares:

$$s_j(\delta(\theta), \theta) = s_j^{obs}, \quad j = 0, 1, \dots, J.$$

The product-level unobserved demand component is then recovered as

$$\xi_j(\theta) = \delta_j(\theta) - X_j \beta.$$

Interpretation. Share matching uses the observed sales data exactly. The remaining identifying content comes from requiring the recovered unobserved quality term to be orthogonal to valid instruments.

3.7 Supply side and marginal costs

The paper assumes static Bertrand-Nash pricing by multiproduct firms. Product j 's log marginal cost is

$$\ln(mc_j) = W_j \tau + \omega_j,$$

where W_j contains observed cost shifters and ω_j is the unobserved cost shock.

Firm f 's variable profit is

$$\Pi_f = M \sum_{j \in J_f} (p_j - mc_j) s_j(p, X; \theta),$$

where M is the number of U.S. households and J_f is the set of products owned by firm f .

The Bertrand first-order condition for product j is

$$s_j(p, X; \theta) + \sum_{r \in J_f} (p_r - mc_r) \frac{\partial s_r(p, X; \theta)}{\partial p_j} = 0.$$

Stacking these first-order conditions gives

$$mc = p - \Delta(p, X; \theta)^{-1} s(p, X; \theta),$$

where Δ is the ownership-adjusted matrix of share derivatives.

Interpretation. Once the demand system is estimated, the supply-side first-order conditions imply marginal costs and markups. The same first-order conditions are used to solve counterfactual prices when minivans are removed.

3.8 GMM objective function

The estimator combines two groups of moments:

$$G(\theta) = \begin{pmatrix} G_1(\theta) \\ G_2(\theta) \end{pmatrix},$$

where G_1 are the standard demand and supply moments and G_2 are the CEX micro moments.

The two-step GMM estimator is

$$\hat{\theta} = \arg \min_{\theta \in \Theta} G(\theta)' \widehat{W} G(\theta).$$

Interpretation. Without G_2 , the model uses prices, characteristics, and shares to infer substitution. With G_2 , the model is additionally penalized when it predicts the wrong demographic composition of buyers.

3.9 Consumer welfare

The paper uses compensating variation to value the minivan. The benchmark is the equilibrium with minivans available. The counterfactual removes minivans and solves for new equilibrium prices of remaining vehicles.

For household i , compensating variation CV_i solves conceptually:

$$V_i(y_i, p^{with}, J^{with}) = V_i(y_i + CV_i, p^{without}, J^{without}).$$

Aggregate consumer surplus change is the sum of simulated CV_i across households.

Interpretation. This measure asks: how much money would the consumer need in the no-minivan world to be as well off as in the world with minivans?

3.10 Producer surplus

For each counterfactual, the model solves new equilibrium prices and computes the change in variable profits:

$$\Delta \Pi_f = \Pi_f(p^{cf}, mc; \hat{\theta}) - \Pi_f(p^0, mc; \hat{\theta}).$$

Total producer surplus change is

$$\Delta \Pi = \sum_f \Delta \Pi_f.$$

Interpretation. Producer surplus effects can be positive for the innovator and negative for rivals. The minivan can create consumer value while reducing industry profits if it cannibalizes existing high-margin products.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the demand system

- Product shares identify mean utilities after the BLP inversion.
- Price sensitivity is identified using instruments because price may be correlated with unobserved product quality.
- Random coefficients are identified from how shares vary with the product choice set and characteristics over time.
- Micro moments sharpen identification by telling the model which demographic groups buy which vehicle types.

Interpretation. The paper's main identification improvement is not a new natural experiment. It is additional moment information that makes the structural model less reliant on weak aggregate variation.

4.2 Price endogeneity

Prices are endogenous because unobserved quality ξ_j likely raises both demand and price. A vehicle with attractive unmeasured features may sell at a higher price, so a naive logit regression would understate price sensitivity.

The paper addresses this with instrumental variables embedded in the nonlinear GMM procedure.

Interpretation. Instrumenting price is essential. The movement from OLS logit to IV logit almost doubles estimated price sensitivity, showing that unobserved quality-price correlation matters.

4.3 Instruments

The paper follows the BLP logic and uses instruments based on:

- observed cost-side variables excluded from demand;
- characteristics of other products made by the same firm;
- characteristics of products made by competing firms.

The economic logic is that equilibrium prices depend on how crowded the product is in characteristics space, both within the same firm and among competitors.

Interpretation. If a product faces many close competitors, optimal price and markup differ from a product in an isolated market niche. Product-space instruments capture this competitive pressure.

4.4 Role of micro moments

The CEX moments are informative because they add restrictions such as:

$$E[\text{new vehicle purchase} \mid \text{income group}], \\ E[\text{family size} \mid \text{minivan purchase}], \quad E[\text{family size} \mid \text{station wagon purchase}],$$

with analogous moments for sport-utilities and full-size vans.

Interpretation. These moments help prevent the model from explaining minivan purchases through extreme unobserved logit tastes. Instead, the model must explain minivan demand through observed demographics and product characteristics.

4.5 Counterfactual assumptions

The welfare and profit calculations rely on several structural assumptions:

- firms price according to static Bertrand-Nash first-order conditions;
- the marginal cost function is stable across the counterfactual;
- removing minivans does not change the characteristics of non-minivan vehicles;
- the set of non-minivan products remains fixed;
- consumer preferences are stable when the choice set changes.

Interpretation. The counterfactual is not a reduced-form causal estimate. It is a model-based welfare exercise. Its credibility depends on whether the estimated demand and supply system produces plausible substitution, prices, and markups.

4.6 Credibility diagnostics

The paper provides several diagnostics that favor the model with micro moments:

- parameter estimates become much more precise when micro moments are added;
- model-predicted income of minivan buyers becomes close to the CEX income of minivan buyers;
- compensating variation no longer depends mainly on extreme logit taste shocks;
- estimated markups are positive and plausible, unlike the OLS and IV logit models that imply negative marginal costs for many vehicles;
- predicted substitution from minivans to station wagons and family vehicles is consistent with descriptive sales patterns.

Interpretation. The key argument is internal consistency: the preferred model matches not only sales shares, but also observed buyer demographics and plausible supply-side markups.

4.7 Limitations

- The CEX vehicle-specific samples are small, especially for full-size vans.
- The CEX buyer moments come from 1987–1992, while the vehicle market sample begins in 1981.
- The model does not observe every household’s full choice set and chosen vehicle at consumer-level scale.
- Welfare estimates depend on the maintained distribution of unobserved tastes and the extreme-value logit component.
- Producer-profit estimates depend on the Bertrand-Nash pricing assumption and on the estimated marginal cost function.

Interpretation. These limitations do not overturn the contribution. They clarify that the paper’s advance is better structural discipline, not perfect experimental identification.

5 Tables and figures

5.1 Table 1: Vehicle characteristics

- **Read.** The table reports sales-weighted characteristics for all vehicles and selected family vehicles during 1981–1993. Minivans are large relative to all vehicles, have relatively high standard air conditioning, and are priced similarly to other family vehicles.
- **Economic message.** Family vehicles occupy a distinct part of product space. Minivans combine size and family functionality with passenger-car-like attributes, making them plausible substitutes for station wagons and large family vehicles.

Table 1: Selected descriptive product facts from Table 1

	All vehicles	Minivans	Station wagons	Sport-utilities
Horsepower/weight	0.39	0.36	0.35–0.37	0.40
Size	0.71	0.87	0.60–0.78	0.83
Air conditioning standard	0.27	0.78	0.00–0.16	0.56
Miles/dollar	24.11	22.84	22.24–26.50	20.20
List price	9,867	10,060	6,864–10,421	10,949

5.2 Table 2: Buyer demographics

- **Read.** New-vehicle buyers are richer than average households. Minivan buyers have the largest family size among the reported vehicle categories: 3.86 household members on average.
- **Economic message.** Minivans are especially attractive to larger families. This is precisely the demographic substitution pattern that the micro moments are designed to identify.

TABLE 1
AVERAGE (Sales-Weighted) CHARACTERISTICS FOR SELECTED VEHICLE TYPES, 1981–93

	ALL VEHICLES		STATION WAGONS			SPORT- UTILITIES	FULL- SIZE VANS
	Mean	Standard Deviation	MINIVANS	$p < \$7,754$	$p \geq \$7,754$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Horsepower/ weight	.39	.07	.36	.35	.37	.40	.30
Length x width x height	.71	.13	.87	.60	.78	.83	1.13
Air conditioning standard	.27	.44	.78	.00	.16	.56	.37
Miles/dollar	24.11	6.55	22.84	26.50	22.24	20.20	17.56
Front wheel drive	.64	.48	.63	.83	.60	.00	.00
List price (1982– 84 CPI \$)	9,867	4,559	10,060	6,864	10,421	10,949	10,321

NOTE.—List price is the base model price. The lowest list price quartile of station wagons is separated out from the top 75 percent.

TABLE 2
AVERAGE CONSUMER CHARACTERISTICS FOR THE UNITED STATES AND SELECTED
SUBPOPULATIONS, 1987–92

	UNITED STATES		PURCHASERS OF				
	Mean	Standard Deviation	New Vehicles	Minivans	Station Wagons	Sport- Utilities	Full-Size Vans
Income	23,728	21,255	36,113	39,476	40,196	41,569	31,164
Family size	2.58	1.53	2.87	3.86	3.17	2.97	3.47
Midage	.55	.49	.64	.78	.73	.74	.65

SOURCE.—Consumer Expenditure Survey.
NOTE.—Income is measured in 1982–84 CPI-adjusted dollars. Family size is the number of household members. Midage is a binary variable for the age of the head of household between 30 and 60 inclusive.

Table 2: Selected buyer demographics from Table 2

	U.S. households	New vehicles	Minivans	Station wagons	SUVs
Income	23,728	36,113	39,476	40,196	41,569
Family size	2.58	2.87	3.86	3.17	2.97
Midage head	0.55	0.64	0.78	0.73	0.74

5.3 Table 3: Minivan and station wagon sales

- **Read.** Minivan share rises from 1.58% in 1984 to 9.93% in 1993. Station wagon share falls from roughly 9–10% before the minivan to about 3% by 1993.
- **Economic message.** The raw sales patterns point strongly toward substitution between minivans and station wagons. A model that treats minivan demand as mostly idiosyncratic will miss this economically important substitution.

Table 3: Family vehicle sales shares: selected years from Table 3

Year	Minivans	Station wagons	SUVs	Full-size vans	Minivan + wagon
1981	0.00	10.48	2.53	1.01	10.48
1984	1.58	8.90	5.51	1.20	10.48
1988	5.97	5.14	7.18	0.69	11.11
1993	9.93	3.02	11.66	0.29	12.95

5.4 Tables 4 and 5: Demand estimates

- **Read.** The paper estimates four demand models: OLS logit, IV logit, random coefficients without micro moments, and random coefficients with CEX micro moments.
- **Read.** Moving from OLS to IV increases estimated price sensitivity, consistent with price endogeneity.
- **Read.** Adding micro moments makes the random-coefficients estimates much more precise. In the model without micro moments, only six of 24 demand-side estimates have Z -statistics above one; with micro moments, 21 of 24 do.

Year	Minivans (1)	Station Wagons (2)	Sport- Utilities (3)	Full-Size Vans (4)	Minivans and Station Wagons (5)	U.S. Auto Sales (Millions) (6)
1981	.00	10.51	.58	.82	10.51	7.58
1982	.00	10.27	.79	1.17	10.27	7.05
1983	.00	10.32	3.51	1.04	10.32	8.48
1984	1.38	8.90	5.51	1.20	10.48	10.66
1985	2.52	7.53	6.11	1.05	9.65	11.87
1986	3.63	6.70	5.75	.85	10.45	12.21
1987	4.86	6.47	6.44	.73	11.55	11.21
1988	5.97	5.14	7.18	.69	11.11	11.76
1989	6.45	4.13	7.47	.61	10.58	11.06
1990	7.55	3.59	7.78	.57	11.54	10.51
1991	8.29	3.05	7.80	.39	11.54	9.75
1992	8.77	3.07	9.33	.39	11.84	10.12
1993	9.55	3.02	11.66	.29	12.05	10.71

Variable	OLS Logit (1)	Instrumental Variable Logit (2)	Random Coefficients (3)	Random Coefficients and Microdata (4)
A. Price Coefficients (α 's)				
α_1	.07 (.01)**	-.13 (.01)**	4.92 (9.78)	7.52 (1.24)**
α_2			11.89 (21.41)	31.15 (4.07)**
α_3			37.92 (18.64)**	54.49 (2.56)**
B. Base Coefficients (β 's)				
Constant	-10.03 (.32)**	-10.04 (.34)**	-12.74 (5.65)**	-15.67 (4.39)**
Horsepower/weight	1.48 (.34)**	3.78 (.44)**	3.40 (9.79)	2.85 (8.16)
Size	3.17 (.26)**	3.25 (.27)**	4.60 (84.64)	4.80 (5.57)*
Air conditioning standard	-20 (.06)**	-21 (.06)**	-1.97 (2.23)	3.88 (2.21)*
Miles/dollar	.18 (.05)**	.05 (.06)**	-.54 (3.80)	-15.79 (4.77)**
Front wheel drive	.32 (.05)**	.15 (.06)**	-5.24 (3.09)	-12.32 (2.36)**
Minivan	.09 (.14)	-.10 (.15)	-4.54 (13.16)	-3.65 (.68)**
Station wagon	-1.12 (.06)**	-1.12 (.07)**	-20.52 (96.17)	-1.51 (.36)**
Sport-utility	-.41 (.09)**	-.61 (.10)**	-5.10 (10.76)	-4.38 (4.41)**
Full-size van	-1.73 (.16)**	-1.89 (.17)**	-28.54 (255.51)	-5.26 (1.30)**
% change GNP	.03 (.01)**	.03 (.01)**	.08 (.02)**	.24 (.02)**

Notes.—Standard errors are in parentheses. A quadratic time trend is included in all specifications.
* *Asymptotic* <1;
** *Asymptotic* <2.

Variable	Uses No Microdata (1)	Uses GEN Microdata (2)
Constant	1.46 (.37)**	5.25 (.72)**
Horsepower/weight	1.48 (14.15)	10 (1.60)**
Size	3.17 (8.69)	4.80 (1.07)
Air conditioning standard	-.20 (.05)*	3.88 (.78)
Miles/dollar	.18 (.04)	-.54 (2.58)
Front wheel drive	.32 (.06)**	-12.32 (4.42)
γ_{10}	.09 (2.62)	-.10 (.14)**
γ_{11}	-.12 (3.43)	-.12 (.28)
γ_{12}	-.41 (5.39)	-.61 (.69)**
γ_{13}	-1.73 (2.84)	-1.89 (.89)**
γ_{14}	.03 (2.23)	.08 (.21)**

Notes.—The OLS and instrumental variable models assume that these random coefficients are zero. Standard errors are in parentheses. A quadratic time trend is included in all specifications. The subscript stands for minivan, so for station wagon, so for sport-utility, and so for full-size passenger van.
* *Asymptotic* <1;
** *Asymptotic* <2.

- **Economic message.** The micro moments are not cosmetic. They materially discipline the parameters most relevant for minivan substitution and welfare.

5.5 Table 6: Supply-side cost estimates

- **Read.** Estimated marginal cost rises with acceleration, weight, fuel efficiency, and air conditioning. The coefficient on log quantity is negative.
- **Economic message.** The supply side produces sensible cost patterns and supports counterfactual pricing calculations. The negative quantity coefficient is consistent with scale economies.

Variable (τ 's)	Parameter Estimate	Standard Error
Constant	1.50	.08
ln(horse power/weight)	.84	.03
ln(weight)	1.28	.04
ln(MPG)	.23	.04
Air conditioning standard	.24	.01
Front wheel drive	.01	.01
Trend	-.01	.01
Japan	.12	.01
Japan $\times$ trend	-.01	.01
Europe	-.47	.03
Europe $\times$ trend	-.01	.01
ln(q)	-.05	.01

TABLE 7
EQUILIBRIUM PRICES WITH AND WITHOUT THE MINIVAN, 1984:
1982-84 CPI-ADJUSTED DOLLARS

	PRICE		% Δ PRICE	
	With Minivan	Without Minivan	Δ PRICE	Δ PRICE
A. Largest Price Decreases on Entry				
GM Oldsmobile Toronado (large sedan)	15,502	15,643	-141	.90
GM Buick Riviera (large sedan)	15,379	15,519	-139	.89
GM Buick Electra (large sedan)	12,843	12,978	-135	1.04
GM Chevrolet Celebrity (station wagon)	8,304	8,431	-127	1.51
Ford Cadillac Eldorado (large sedan)	19,578	19,704	-126	.64
Ford Cadillac Seville (large sedan)	21,625	21,749	-125	.57
GM Pontiac 6000 (station wagon)	9,273	9,397	-123	1.31
GM Oldsmobile Ciera (station wagon)	9,591	9,714	-123	1.27
GM Buick Century (station wagon)	8,935	9,056	-121	1.34
GM Oldsmobile Firenza (station wagon)	7,595	7,699	-104	1.35
B. Largest Price Increases on Entry				
Chrysler LeBaron (station wagon)	9,869	9,572	297	3.10
Volkswagen Quattro (station wagon)	13,263	13,079	184	1.41
Chrysler (Dodge) Aries K (station wagon)	7,829	7,659	170	2.22
AMC Eagle (station wagon)	10,178	10,069	109	1.08

NOTE.—Equilibrium prices without minivans are estimated using the model with microdata and Bertrand-Nash first-order conditions. Bertrand-Nash pricing with random coefficients does not a priori determine signs of firm-specific price changes.

5.6 Table 7: Price changes from minivan entry

- **Read.** When minivans enter in 1984, the model predicts price decreases for many large sedans and station wagons. The largest dollar price decreases are around \$100–\$150.
- **Economic message.** Non-minivan buyers benefit from greater price competition. This channel accounts for almost 43% of total consumer benefits in 1984.

5.7 Table 8 and Figure 1: Compensating variation

TABLE 8
AVERAGE COMPENSATING VARIATION CONDITIONAL ON MINIVAN PURCHASE, 1984:
1982–84 CPI-ADJUSTED DOLLARS

	OLS Logit	Instrumental Variable Logit	Random Coefficients	Random Coefficients and Microdata
Compensating variation:				
Median	9,573	5,130	1,217	783
Mean	13,652	7,414	3,171	1,247
Welfare change from difference in:				
Observed characteristics ($\delta + \mu_o$)	-81,469	-44,249	-820	851
Logit Error (ϵ_μ)	95,121	51,663	3,991	396
Income of minivan purchasers:				
Estimate from model	23,728	23,728	99,018	36,091
Difference from actual (CEX)	-15,748	-15,748	59,542	-3,385

NOTE.—Compensating variation is evaluated at equilibrium prices without minivans. Decomposition of compensation is the average difference in the value of observed and unobserved characteristics between first and second choices. For logit models, the purchase decision is independent of income, so mean purchaser income is mean U.S. household income.

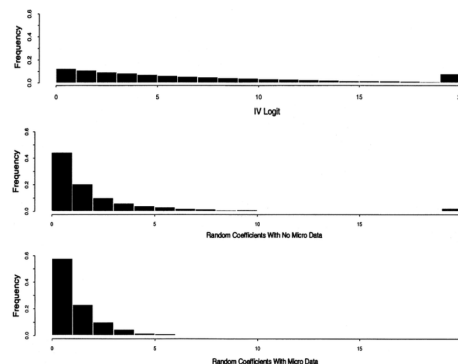


FIG. 1.—Compensating variation for 1984 minivan purchasers: thousands of 1982–84 CPI-adjusted dollars

- **Read.** Average compensating variation for 1984 minivan buyers is \$13,652 under OLS logit, \$7,414 under IV logit, \$3,171 under random coefficients without micro moments, and \$1,247 under random coefficients with micro moments.
- **Read.** In the preferred model, the observed-characteristics component is positive and the logit-error component is much smaller than in the other models.
- **Economic message.** The full model reduces the extreme willingness-to-pay tail. Welfare measurement becomes less dependent on an unexplained idiosyncratic taste shock.

Table 4: Compensating variation for 1984 minivan purchasers from Table 8

	OLS logit	IV logit	RC, no micro	RC, micro
Median CV	9,573	5,130	1,217	783
Mean CV	13,652	7,414	3,171	1,247
Observed-characteristics component	-81,469	-44,249	-820	851
Logit-error component	95,121	51,663	3,991	396
Predicted minivan-buyer income	23,728	23,728	99,018	36,091

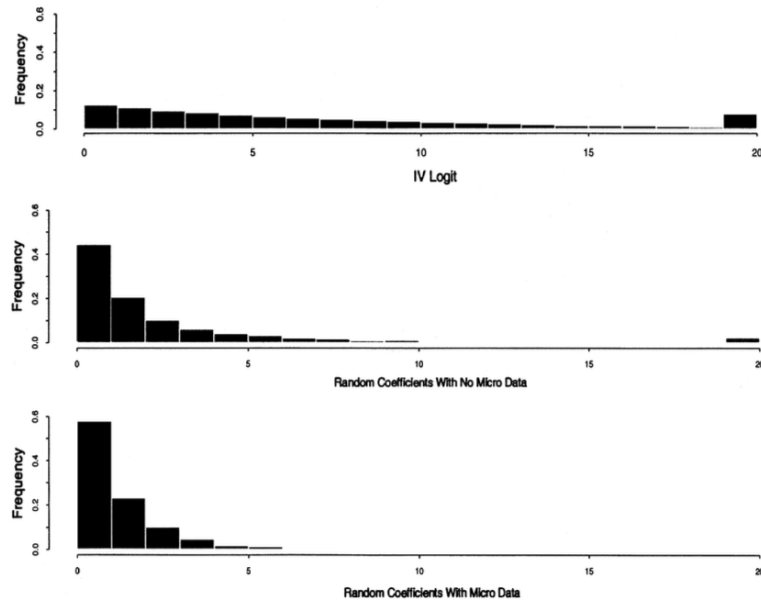


FIG. 1.—Compensating variation for 1984 minivan purchasers: thousands of 1982–84 CPI-adjusted dollars

Figure 1: Distribution of compensating variation for 1984 minivan purchasers. The full model with micro moments greatly reduces the extreme right tail.

5.8 Table 9: Markups

- **Read.** OLS and IV logit imply many negative marginal costs because implied markups are too large. The random-coefficients model with micro moments implies an average dollar markup of \$1,753 and an average percentage markup of 16.7%.
- **Economic message.** A demand model that mismeasures substitution also mismeasures markups and marginal costs. The preferred model gives more plausible supply-side implications.

Table 5: Markup diagnostics from Table 9

	OLS logit	IV logit	RC, no micro	RC, micro
Mean dollar markup	13,904	7,551	4,017	1,753
Mean percent markup	—	—	40.7%	16.7%
Negative estimated marginal costs	73.7%	22.6%	0%	0%

5.9 Tables 10–12: Producer surplus and first-mover advantage

- **Read.** Minivans earn higher percentage markups than station wagons and sport-utility vehicles during the early years.

TABLE 9
IMPLIED MARKUPS DERIVED FROM DEMAND-SIDE ESTIMATES AND BERTRAND-NASH
PRICING ASSUMPTION, 1981-93 (2,407 Models)

STATISTIC	OLS LOGIT (1)	INSTRUMENTAL VARIABLE LOGIT (2)	RANDOM COEFFICIENTS AND MICRODATA		(5)	(6)
			(3)	(4)		
Median	\$13,834	\$7,513	\$2,593	36.7%	\$1,439	15.0%
Mean	\$13,904	\$7,551	\$4,017	40.7%	\$1,753	16.7%
10%	\$13,647	\$7,413	\$1,628	27.8%	\$819	11.2%
90%	\$14,297	\$7,765	\$8,357	62.6%	\$2,856	24.8%
Standard deviation	\$257	\$140	\$4,089	14.0%	\$1,229	6.2%
Estimated marginal costs that are negative	73.7%	22.6%	0%	0%		

NOTE.—Percentage markups are estimated markups divided by observed prices. They are not reported for instrumental variable and OLS logits because the estimated marginal cost is negative for many vehicles. Dollars are 1982-84 CPI adjusted.

TABLE 10
AVERAGE (Sales-Weighted) MARKUP/PRICE AND PRICE, 1983-87

YEAR	MINIVANS	STATION WAGONS	SPORT-UTILITY VEHICLES	ALL OTHER VEHICLES
1983	...	...	18.27%	\$8,230
1984	19.19%	\$8,722	17.83%	\$8,400
1985	19.43%	\$8,655	16.81%	\$8,412
1986	17.33%	\$8,898	15.82%	\$9,164
1987	16.31%	\$9,527	13.86%	\$9,620

NOTE.—Percentage markups are estimated markups divided by observed prices. Dollars are 1982-84 CPI adjusted.

- **Read.** Chrysler gains large variable profits from introducing the minivan: \$202.5 million in 1984 and \$504.1 million in 1988.
- **Read.** Ford and GM lose profits from Chrysler's innovation because minivans cannibalize their existing product lines.
- **Read.** Entry by Ford and GM minivans dissipates part of Chrysler's first-mover advantage, reducing Chrysler profits by an estimated \$126 million over 1985-1988.
- **Economic message.** The minivan produced a classic innovation pattern: large temporary private returns for the innovator, losses for rivals, and gradual profit erosion through imitation.

Table 6: Selected producer-profit effects from Tables 11 and 12

Year	Chrysler profit effect	Ford profit effect	GM profit effect	Chrysler loss from imitators
1984	202.5	-31.8	-155.8	-
1985	259.1	-37.4	-171.0	-6.1
1986	201.1	54.7	-119.9	-22.7
1987	346.1	-22.8	-174.5	-42.4
1988	504.1	-24.7	-235.4	-55.7

TABLE 12
CHRYSLER'S PROFIT DISSIPATION WITH
ENTRY OF FORD AND GM MINIVANS

YEAR	CHANGE IN TOTAL VARIABLE PROFITS	
1985	-\$6.06	-.16%
1986	-\$22.72	-1.99%
1987	-\$42.35	-2.25%
1988	-\$55.68	-2.63%

NOTE.—These profit changes are computed using the model (see the text).

TABLE 11
CHANGE IN INDUSTRY AND BIG THREE TOTAL VARIABLE PROFITS WITH THE ADVENT OF MINIVANS

YEAR	INDUSTRY	CHRYSLER	FORD	GM
1984	-21%	\$202.5	14.38%	-\$31.8
1985	-13%	\$259.1	13.99%	-\$37.4
1986	-14%	\$201.1	12.43%	-\$54.7
1987	-17%	\$346.1	23.27%	-\$22.8
1988	-25%	\$504.1	32.50%	-\$24.7

NOTE.—Dollar figures are given in millions. The numbers are computed using the model to estimate profits both with minivans in the market and with minivans removed from the market (see Sec. V).

TABLE 13
CHANGE IN U.S. WELFARE FROM THE MINIVAN INNOVATION, 1984-88 (\$ Millions)

Year	Compensating Variation	Change in Producer Profits	Welfare Change
1984	\$67.29	-36.68	\$30.61
1985	\$25.04	-25.07	\$99.97
1986	\$39.53	27.30	\$67.23
1987	\$96.59	29.75	\$66.84
1988	\$75.70	110.24	\$85.94
Total	\$284.55	105.54	\$219.09

NOTE.—Computations were done using 1982-84 CPI-adjusted dollars.

5.10 Table 13: Total welfare

- **Read.** Over 1984-1988, consumer surplus rises by about \$2.805 billion. Producer profits rise by about \$105.5 million. Total welfare rises by about \$2.91 billion.
- **Economic message.** Most of the value of the minivan accrues to consumers, not the innovator. The private return to innovation is large, but much smaller than the social return.

Table 7: Welfare change from the minivan innovation, 1984–1988

Year	Consumer surplus	Producer profits	Total welfare
1984	367.29	−36.68	330.61
1985	625.04	−25.07	599.97
1986	439.93	27.30	467.23
1987	596.59	29.75	626.34
1988	775.70	110.24	885.94
Total	2,804.55	105.54	2,910.09

6 Short summary

- Petrin studies the consumer and producer welfare effects of the minivan introduction in the U.S. automobile market.
- The paper’s methodological contribution is to augment market-level BLP demand estimation with aggregate micro moments from the CEX.
- These micro moments describe which demographic groups buy which vehicle types, especially family size among minivan and station wagon buyers and purchase probabilities by income group.
- The preferred random-coefficients model with micro moments produces more precise demand estimates, more plausible substitution patterns, and more reasonable markups.
- Welfare estimates without micro moments are too large because they rely heavily on the idiosyncratic logit taste shock.
- In the preferred model, average compensating variation for 1984 minivan buyers is \$1,247, much lower than in simpler logit models.
- Consumers gain both because minivans become available and because the new product increases price competition among substitutes.
- Chrysler earns large first-mover profits, but Ford and GM lose profits as minivans cannibalize existing family vehicles.
- Over 1984–1988, total welfare gains are about \$2.91 billion, most of which is consumer surplus.

7 Conclusion

7.1 Core conclusion

The paper establishes that the estimated value of a new differentiated product depends critically on the credibility of substitution patterns. The minivan generated large welfare gains, but simpler models substantially overstate those gains because they explain consumer choices through extreme idiosyncratic logit tastes. Adding aggregate micro moments makes the model’s substitution patterns more realistic and sharply changes welfare estimates.

7.2 Economic intuition

A minivan is not valuable to all households equally. It is especially valuable to households with children and larger families, and it is close to station wagons and other family vehicles. If the model does not know this, it may infer that minivan buyers have enormous unobserved tastes for minivans. That creates implausibly large compensating variation. The CEX moments force the model to connect minivan demand to observed demographics, which produces more reasonable willingness-to-pay and markup estimates.

7.3 Takeaway

Petrin's paper is both a new-product welfare exercise and a methodological lesson. New products can create social benefits much larger than private profits, but measuring those benefits requires demand systems that capture who substitutes to what. Aggregate demographic moments are a practical way to improve structural demand estimation when full consumer-level choice data are unavailable.